Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829 · LINE solaress1
www.solaress.org · www.โซลาร์เอสส์.com

TECHNICAL NOTE · INVERTER SETTINGS

Reading a hybrid inverter's settings: a UK grid code on a Thai roof

Case: 30 kWp three-phase hybrid at a resort in Chiang Rai, no battery yet · settings read remotely in SolisCloud · Rev 01 · 30 October 2026

G59/3
grid code set in the inverter: the UK standard, not a Thai one

55 kW
export limit, higher than the 30 kWp array, so it never applies

Enable
grid charging allowed, with no battery fitted

ON
unbalanced output, right for mixed single-phase loads

Settings decide how a hybrid inverter behaves long after the installer has left. We read every setting of one inverter remotely, without changing anything. Most were sensible, one needs correcting, and two are worth knowing about before a battery is added.

1 What we found, setting by setting

Setting	Value	What it means	Action
Storage Mode	Self Use	Solar serves the load first; this is normal.	None
Allow Export / Max Export Power	Enable / 55 kW	Export is allowed and the limit sits above the array size, so it never limits anything.	Match it to the export agreement with PEA.
Export Calibration	−256 W	A small offset that keeps a little grid import as a safety margin.	None
Unbalance Output	ON	Each phase is fed according to its own load, which suits resorts with single-phase loads.	None
Allow Grid Charge	Enable	No effect without a battery, but a future battery would charge from the grid at any time.	Decide deliberately before a battery is added.
Battery Reserve	OFF (80%)	No battery is connected, so it has no effect.	Set when a battery is added.
Grid Peak Shaving	OFF	Not used.	None
Grid Code	G59/3	The UK grid code, with UK voltage and frequency trip limits.	Change to the Thai code the inverter was approved under.

2 Why the grid code matters

The grid code tells the inverter when to disconnect because the grid voltage or frequency is out of range, and how fast. In Thailand, inverters connected to the PEA or MEA network are tested against those authorities' grid-connection regulations, which build on IEC 61727 and IEC 62116. The inverter must run with the setting it was approved under. The trip limits read from this inverter:

Limit	Value	Meaning
Over-voltage, stage 1 / 2	262.2 V (1 s) / 273.7 V (0.5 s)	disconnects if the voltage stays high
Under-voltage, stage 1 / 2	200.1 V (2.5 s) / 184 V (0.5 s)	disconnects on low voltage
Over-frequency, stage 1 / 2	51.5 Hz (90 s) / 52 Hz (0.5 s)	
Under-frequency, stage 1 / 2	47.5 Hz (20 s) / 47 Hz (0.5 s)	
Reconnect window	200.1–262.2 V, from 47.5 Hz	range in which it starts again

What to do. The installer should select the Thai grid code that matches the inverter's PEA approval, note the settings before and after, and record the change in the site log. A grid code is a safety setting: change it only on-site or with the installer's access, never to "stop trips" without understanding why they happen. If the inverter trips on high voltage in the afternoon, the cause is usually the supply, which may need PEA, not a wider setting.

3 Settings to read on every hybrid installation

1. Grid code: the Thai code the inverter was approved under (PEA or MEA).
2. Storage mode and work mode: Self Use, Feed-in Priority, or time of use as agreed with the owner.
3. Export: allowed or zero export, and the limit, matching the agreement with PEA.
4. Unbalanced output on three-phase sites with single-phase loads.
5. Grid charging: on only if charging from the grid is intended (for example cheap off-peak power).
6. Battery: type, charge and discharge current, minimum SOC and reserve for outages.
7. Meter or CT direction and phase order; check that the live data makes sense.
8. Record every value in the site log, so changes can be traced later.

In one sentence: a quick remote read of the settings found one real problem, a UK grid code on a Thai installation, plus one switch to decide before adding a battery. It costs nothing and should be part of every commissioning and annual check.

References

1. SolisCloud — Inverter Control page of a 30 kWp resort plant (Storage Mode, Battery Setting, Grid Port), read 30 October 2026
   Monitoring data
2. SMA — Certificate of conformity for Thailand: IEC 61727, IEC 62116 and the MEA / PEA grid-connected inverter regulations
   https://files.sma.de/downloads/ZE_IEC_61727_MEA_PEA_STPxx000TL-30_en_10.pdf
3. KMUTT — Difference of grid code for PV inverters in Thailand and other countries
   https://kirim.kmutt.ac.th/converis/portal/detail/Publication/1277898430?lang=en_GB
4. Solis — Grid standards for Solis inverters (how grid codes are selected and customised)
   https://usservice.solisinverters.com/support/solutions/articles/73000560465-grid-standards-for-solis-inverters

Prepared by Solar ESS for general information, 30 October 2026. Settings were read remotely and not changed. Grid-code changes must be made by the installer in line with PEA or MEA requirements.